Problem Recognition and the Limits of Commons Theory

Peter Pasholk

PLS 4793: Environmental Politics and Policy

Dr. Paul Danyi

May 3, 2026

1. Introduction

In 1962, the publication of *Silent Spring* by Rachel Carson marked a turning point in environmental policy. By effectively creating public awareness of the ecological and human health impacts associated with the widespread use of DDT on other synthetic pesticides, Carson's contributions ultimately lead to the federal ban of DDT in the United States (Carson, 1962; PBS, 2017). While the policy outcomes of DDT and the story behind the publication and reception of *Silent Spring* is well documented, less attention has been given to how this transformation occurred in theoretical terms. For this case in particular, a foundational question for commons governance is raised: how do environmental problems become shared collective action problems, or commons dilemmas, in the first place?

Common pool resources, or commons, are shared resources, like air or water, that are difficult to exclude users from, but are finite and rivalrous in supply (Hardin, 1968). This paper will situate Carson's work in relation to the two dominant frameworks in commons theory; Garrett Hardin's *Tragedy of the Commons* and Elinor Ostrom's theory of polycentric governance, and ask how, if at all, Carson fits within them.

Hardin predicts that individuals will act in rational self-interest and inevitably overexploit shared resources without privatization or privatization. Ostrom demonstrates that communities can sustainably govern these shared resources through institutional design principles. Both frameworks make a critical assumption that commons problems are already recognized by relevant actors. Further, neither theory adequately explains how diffuse, invisible, or contested environmental harms become collective action problems that demand governance in the first place.

1.1 Contribution:

This paper argues that Carson's *Silent Spring* reveals a missing pre-institutional layer in commons theory that is centered on problem recognition, communication, and normative transformation. By analyzing the DDT ban through the perspective of commons theory, the paper identifies a missing polycentric activation stage between public problem recognition and institutional rule formation that corrects Hardin and expands Ostrom while acting as a bridge between the two ideas.

By identifying and theorizing this missing layer, this paper makes three contributions to commons theory. First, it extends commons theory by incorporating pre-institution processes of communication and activation. Second, it reconciles key tensions between Hardin and Ostrom by positioning Carson as a bridge between problem identification and institutional design. Third, it offers a revised, multi-stage framework for collective actions that better account for large-scale environmental governance in contemporary contexts.

2. Theoretical Framework

Commons theory provides a central framework for understanding environmental governance, especially in contexts where resources are shared, competitive, and nonexcludable. The two foundational articles of commons theory, Garrett Hardin's "Tragedy of the Commons" and Elinor Ostrom's theory of polycentric governance, offer contrasting explanations for how resources are managed and under what conditions collective action succeeds or fails. While these frameworks have shaped decades of environmental policy, both make assumptions that limit their ability to explain how environmental problems emerge as objects of governance in the first place.

2.1. Hardin and the Tragedy of the Commons

In the influential 1968 essay, Garrett Hardin argues that individuals, acting rationally and in their own self-interests, will inevitably exploit, and overexploit, shared resources. Resource users individually benefit from using common-pool resources while the costs of degradation are distributed across all users, leading to an inevitable collective depletion of the resource. Hardin argues that the tragedy is this fate is inescapable in the absence of external constraints.

Hardin's framework makes several core assumptions outlined in Table 1 in the appendix. Most notable is that individuals are driven by rational self-interest and will maximize personal gain at the expense of collective outcomes. In the tragedy, cooperation fails to emerge because actors lack trust and shared norms, making self-restraint illogical. Even when actors understand the problem, Hardin assumes that incentives to overuse will dominate behavior, creating a commons dilemma. Given these assumptions, Hardin concludes that there are only two viable solutions: mutually accepted coercion: the privatization of the resource or coercive regulation imposed by an external, central authority.

2.2. Ostrom and Polycentric Commons Governance

In contrast to Hardin's pessimistic conclusion, Elinor Ostrom's work demonstrates that communities are capable of sustainably managing common-pool resources without resorting to privatization or top-down control. She simultaneously counters Hardin and extends commons theory by proposing a third solution; a set of collective action design principles for the successful governance of common pool resources outlined in Table 2 in the appendix. Central to this framework is the concept of polycentric governance, where multiple layers of overlapping authority operate at different scales. Instead of relying on centralized control, commons management emerges through interactions between local users, institutions, and regulatory systems. Instead of assuming people will act in their own self-interest, Ostrom emphasizes trust, communication, and norms can establish cooperation.

Despite its strengths, Ostrom's framework has important limitations. Her design principles assume that resource users have already recognized the existence of a commons dilemma and share the same understanding of the problem. There are additional assumptions of an existing level of communication and mutual awareness that are not always present, especially on large scales or diffuse issues. Further, Ostrom's focus is primarily on small- to medium-scale applications, leaving a gap about how her framework applies to national or global environmental challenges.

2.3. Identifying the Theoretical Gap

Hardin and Ostrom's commons theories defines their own boundaries: the former emphasizes why specific structural constraints inevitably lead to collective action failure and the latter highlights specific institutional and social conditions under which cooperation can emerge. Both frameworks share a critical blind spot to account for how environmental problems become recognized as commons dilemmas in the first place. They are also assuming the process which public awareness and mobilization, scientific knowledge and moral concern translate into successful collective action has infrastructure to occur in the first place and will happen natrually.

This gap is especially significant in cases where the environmental harms are diffuse, harm is delayed, or the impact is unevenly distributed. In contexts where the problem is not initially recognized, significantly complex, or if the impact is contested, the emergence of a commons dilemma might not be naturally recognized and need to be deliberately constructed through study, communication, and mobilization.

The case of *Silent Spring* provides a credible lens for examining this gap in commons frameworks. This paper will demonstrate that Carson's work operates not within current frameworks, but prior to them, shaping the conditions in which collective action, and ultimately governance, become possible. By incorporating *Silent Spring* into commons institutions, the following sections will seek to extend commons theory beyond its current limits and provide a more complete, larger-scale account of environmental collective action.

3. Methodology and Research Design

This paper conducts a qualitative, theory-driven study to investigate how outcomes in this DDT case fit, or do not fit, into existing commons theory. The outcomes of the DDT case were compared to Hardin's core assumptions and Ostrom's design principles, presented in Tables 1 and 2. These tables demonstrate that Carson does not meaningfully fit *within* existing commons theory because both Hardin and Ostrom begin at a point where certain preconditions are already in place. Using the insights of these comparisons, a layered analytical framework was constructed to move backward from institutional outcomes and situate how Carson fits *outside* of these two frameworks. The results of these studies identify gaps between Hardin's and Ostrom's arguments, especially their shared assumptions that commons dilemmas are already recognized by resource users and addressing this limitation by theorizing the conditions under which environmental problems become governable.

After seeing how Carson is situated within commons theorists, this study will reinterpret commons theory and solutions for governing the commons through the DDT case and Carson's contributions. This will propose a reconstructed framework for commons governance. Before concluding, this paper will explore what Carson specifically did that made *Silent Spring* successful and the limits of this new framework by comparing similar, but modern, communications and their outcomes.

Silent Spring is an ideal case study because it was a high impact policy shift with a clear before and after where the commons dilemma was not recognized. In fact, scientists praised pesticides as a solution to hunger and disease while the public was unaware of the consequences of unmonitored pesticide usage (Carson, 1962). There is a clear timeline between Carson publishing *Silent Spring* and the final institutional response that is very well documented (PBS, 2017). It is also possible to assess and incorporate Carson's unique communication style and how the distribution of *Silent Spring* helped shaped public understanding into the theories this paper studies.

At the same time, this study has several limitations. As a single-case analysis, its findings are not intended to generalize broad environmental contexts, but to refine current, and generate new, theoretical insights. The focus is on the United States and may not work in international contexts. Further, there is inherent bias from evaluating *Silent Spring* in hindsight and frameworks built from it may struggle to address contemporary challenges. Despite these limitations, this case provides a clear and influential example of how large-scale problem recognition and communication is an assumed but critical element of commons theory.

4. Case study on Carson and DDT

Before *Silent Spring* was published, the widespread use of synthetic pesticides like DDT was unanimously seen as a public good (Carson, 1962). Following World War II, chemical pesticides were celebrated as technological breakthroughs. Farmers were able to grow crops at historic levels of production and there were new solutions for controlling diseases like malaria. These chemicals were promoted with minimal scrutiny and used without regulation throughout the country (Carson, 1962; PBS, 2017). Because of this perspective, farmers were overusing pesticides. While sometimes the consequences were noticed locally but were not interpreted as a broader problem. Within this context, there is no commons dilemma to solve as pesticides known to be a solution, not a risk.

Carson and *Silent Spring* reframed pesticides as a diffuse and interconnected ecological threat. She argued, using a wide range of scientific studies, that chemicals like DDT accumulate in ecosystems, move through food chains, and produce unintended consequences to both wildlife and humans (Carson, 1962). Pesticide use and exposure was reframed as a commons dilemma. The agricultural section and chemical companies benefited from pesticide use while the harms were fragmented and spread throughout whole regions and populations.

The first, and most critical observation in this case is not what Carson communicated, but how. The book opens with the famous “Fable for Tomorrow,” a fictional town where environmental damage leads to the collapse of life in a poetic and sincere tone that could also be used to tell a fairytale. It was a blending of storytelling and science that allowed readers to make the connection between complex ecological relationships, isolated observations, and unmanaged risks of synthetic pesticide use.

The medium Carson used to propose and defend her arguments is also critical to understand. Prior to its release as a book, chapters of *Silent Spring* were serialized in *The New Yorker*. This essentially rolled the book out to a large audience, chapter-by-chapter, over several weeks. The expanded reach from this serialization cannot be understated and audiences were hooked (PBS, 2017). When defending her work on television, Carson presented herself in a quiet, calm demeanor where she did not tell the public what to do, only that we need to take a serious look at indiscriminate use (Carson, 1962). She said pesticides need to be regulated, not banned, and that there are risks to thoughtless use, not that people are wrong for using them. In many ways, she asked, “is this the right way?” and did say, “do not do this.” Opponents found it challenging to disagree with reasonableness and her call to action was open-ended enough for subject matter experts to take over the process.

The response to *Silent Spring* was a dynamic and contested process where industry groups mounted an aggressive counter campaign to discredit Carson (Carson, 1962; PBS, 2017). At the same time, the book galvanized public concern, prompted widespread media attention, and demanded debate within the scientific community begin immediately. The entire nation was swept up in a debate not only about pesticides, but about government priorities and transparency. This case is the marked shift from fragmented, isolated awareness to a universally identified collective action problem with shared risks.

Ultimately, this process culminated in the formal federal recognition of the issue at the highest levels of government. In 1963, President John F. Kennedy publicly acknowledged the concerns raised by *Silent Spring* and directed the Presidential Science Advisory Committee to conduct a comprehensive review of pesticide use (Carson, 1962; PBS, 2017). The report validated Carson and led to the formation of the United States Environmental Protection Agency and the banning of DDT in 1972 (PBS, 2017). Unfortunately, Carson did not live to see the outcoming, passing away in 1964 to cancer.

What Carson did was trigger a multi-layered, polycentric activation. Multiple actors became involved in evaluating, contesting, and responding to the claims made by *Silent Spring*. The issue moved quickly from debate to politics, starting a series of regulatory investigations and actions that reshaped U.S. environmental policy and the eventual ban of DDT. While the controversy and debate around *Silent Spring* contributed to the emergence of the modern environmental movement, it is important to understand that it did not happen spontaneously like existing commons theory describes. This distinction is critical for the following analysis that highlights the limits of existing frameworks while positioning Carson within them.

5. Analysis on Carson and the Limits of Commons Theory

5.1. Carson Within Existing Commons Frameworks

The comparison analysis presented in Tables 1 and 2 shows consistently that *Silent Spring* does not meaningfully fit into Hardin's assumptions nor Ostrom's design principles. When broken out granularly, elements of the DDT case partially support these theories, neither framework accounts for the drivers of this case study. Instead, Carson exposes how communication interventions, problem recognition, and public mobilization take place outside the existing commons theory framework.

Across Hardin's Framework, the comparison shows a pattern of partial confirmation at the level of diagnosis by a breakdown at the level of assumptions about behavior and systematic outcomes. The DDT case aligns with Hardin's premise that actors behave according to what is in their rational self-interest. Farmers and chemical producers adopted pesticides because they increased yields and profits. Where Hardin's proposed solution of coercive regulation appears to hold, the pathway in the DDT case differs. Regulation emerges not as an externally imposed necessity, but as the product of democratic pressure and legitimacy-building. With that in mind, Carson corrects Hardin that structural risks can be addressed and the dilemma can be interrupted through social and political processes.

A similar pattern emerges from Ostrom's framework. Certain design principles, such as nested governance and monitoring, find support in the DDT case. The policy response unfolded across multiple layers of governance and Carson's synthesis of evidence was a kind of monitoring that made the harm visible. However, most of Ostrom's principles are a weak fit with the DDT case. Carson describes a fundamental issue with pesticides is the clear lack of boundaries, making it especially challenging to define a community of resource users. What is interesting is that mechanisms such as graduated sanctions and accessible conflict resolutions were absent. In the DDT case, these steps were replaced by conflict between actors in the form of debates, news reporting, and legal disputes. Instead of local users organizing, as Ostrom predicted, a broad public mobilization occurred. The findings in the DDT case suggest that Ostrom's framework does not necessarily occur outside of small- and medium-sized problems and struggles to account for large-scale, contested environmental problems.

It is not that one framework performs better than the other, but that both share common limitations. Hardin and Ostrom begin their analyses at a point where the commons dilemma is already recognized, and actors understand the stakes and share a good degree of knowledge. The DDT case demonstrates that these conditions cannot be assumed. Prior to Carson's intervention, pesticide use was perceived as a collective good. The central dynamic of the case is therefore not the management of common pool resources, but the transformation of an unrecognized condition into a shared problem requiring commons governance.

5.1. Carson Outside of Existing Commons Frameworks

Because Carson cannot be meaningfully situated within either framework, an analysis was needed to step outside of them. The four-layer analysis framework in Figure 1 used in this paper emerges directly from the comparative evaluation of Hardin's assumptions and Ostrom's design principles presented in Tables 1 and 2. The analysis layers are designed to move backward from institutional outcomes to the deeper foundations outside of traditional frameworks that make those outcomes possible.

The layers in this analytical framework comes from the limits directly identified by Hardin and Ostrom; institutional arrangements, cooperation dynamics, and structural conditions in which resources are managed. However, layers were added to address the framework's assumptions that the commons dilemma is already recognized. Each layer is intended to correspond to a distinct stage in the formation of collective action while mapping in Carson's contribution to fill gaps within existing theory. Working backwards, from the commons being managed to the problem being first identified, we can now place Carson and the DDT case in relation to Hardin and Ostrom in commons theory.

Figure 1.

Layer 1: Institutions: Formal and informal rules, monitoring, and enforcement. ○ Carson does not fit and corrects institutional rule forming's place in commons theory. ○ Coercion emerges *after* public mobilization and legitimacy-building. ○ The DDT ban is top down, adversarial, not a self-governed commons. ○ Institutions and polycentric governance are not the starting point; they are the endpoint.	**Solution Implemented**
Layer 2: Collective-action dynamics: The internal social variables of trust, norms, reciprocity that influence cooperation. ○ Carson fits partially. She builds norms and shared understanding that make collective action possible. ○ Hardin and Carson agree about structural risks of commons dilemmas but disagrees on the outcome Hardin proposed, which was that cooperation is structurally unlikely.	**Solution Development**
Layer 3: Problem recognition: The process in which symptoms are identified as a threat. ○ Carson fits strongly. She reveals the commons, names the threat, and creates a common fate. ○ The unspoken foundation explains how a situation becomes a problem at all. ○ Ostrom assumes this layer is pre-existing in design principles. Hardin assumes actors understand the dilemma. ○ Carson extends Ostrom and corrects Hardin.	**Problem Recognition**

○ Silent Spring is foundation work that reveals interdependencies, names threat, makes diffuse harm visible.	
Layer 4: Cultural and communicative foundations: The moral frameworks and narratives that make commons visible and meaningful. ○ Carson is essential. She constructs the very conditions under which a commons can be seen and therefore governed. "Fable for tomorrow," moral framing ○ This is missing from commons theory. Hardin dismisses communication, Ostrom assumes it. ○ Hardin misses that communication is a precondition for governance.	**Norms Established**

Four-Layer Analysis of Commons Theory

Using this analysis to show an untheorized foundation within commons theory, there is room to explore why commons dilemmas are not inevitable and how to address them on larger scales. These layers of problem recognition and communication are implied in both Hardin and Ostrom, yet they do not explain how actors arrive at shared understandings or how values are shaped and activated.

This study shows that Carson is not situated as a participant in commons governance, but as an agent who creates the epistemic, moral, and narrative conditions that Hardin and Ostrom assume in their arguments. By reconstructing Ostrom's framework and Hardin's assumptions into these layers, we have an analytical basis for reinterpreting commons governance theory through Carson's contribution.

6. Reinterpreting Commons Governance Through Carson

6.1. Reinterpreting Core Assumptions of Collective Action

The limitations of commons theory are not simply empirical gaps, they stem from foundation assumptions made by the authors on human behavior, information, and governance. Reinterpreting these assumptions through the DDT case shows that both Hardin and Ostrom begin their analysis late in the chain of collective action events.

In Hardin's framework, he assumes that information and appeals to conscience are irrelevant to the outcomes. Individuals knowingly overexploit shared resources because incentives outweigh other factors. The DDT case specifically contradicts this premise. Individual perceptions of risks and responsibilities were fundamentally reshaped after the publication of *Silent Spring*. Her combination of narrative writing, scientific communication, and moral framing not only informed the public of the risks, but it also transformed it into a shared problem. This is how Carson shows that communication and norm forming is a precursor to collective action.

Further, *Silent Spring* counters Hardin's assumption that the tragedy is inevitable. Public awareness created the pathway to widespread engagement between multiple levels of actors. When citizens, scientists, journalists, and advocacy groups all shared the same concerns, trust and norms emerged and created regulatory pressure. It is not that there is an inevitable structural failure of collective action, but it is contingent on how shared-fate problems are understood or communicated.

Even though the command-and-control outcome aligns with Hardin's proposed solutions for governing the commons, the DDT case offers an important correction; coercion did not proceed mobilization or collective action, it followed it. Regulation emerged from mobilization, scientific validation, and political contestation, rather than being imposed as a purely external constraint. In some ways, Hardin models the endpoint without accounting for the process that makes it politically and socially possible. Finally, Hardin's determination to treat commons dilemmas apolitically, he overlooks the role of power, legitimacy, and communication that the DDT case sharply corrects.

While Ostrom's framework corrects many of Hardin's limitations on its own, it presupposes a key condition to the design principles that Carson helps explain. Ostrom's design principles assume the existence of problem recognition, communication, and a baseline of trust. The DDT case shows how essential these conditions are, especially at larger scales where harms are diffuse and contested. In fact, opinion on pesticides was entirely the opposite of problematic until *Silent Spring* generated these exact preconditions necessary for collective action to occur on a national scale the Ostrom's framework does not typically work at.

Together, these insights point to a missing stage in commons theory: a process through which problems are recognized, norms are constructed, and multiple actors become activated prior to formal governance. Neither Hardin nor Ostrom explicitly theorize this polycentric activation stage, but it is essential for explaining how environmental actions for managing commons become possible. Carson did not necessarily communicate the need for the public to take collective action in *Silent Spring* or her communications during its publication; she acted as a trigger that can be explored as a theoretical construct explaining why collective action occurred. The DDT case suggests this process is not always linear or cooperative, and accounts for the contestation, persuasion, and interaction of diverse actors across domains.

6.2 Carson as the Bridge in Commons Theory

Positioning Carson within commons theory requires moving beyond “fit” and reconsidering the structure of the theory itself. Hardin and Ostrom can be seen as addressing different moments within the broader processes of collective action. Hardin diagnoses the structural conditions in which commons dilemmas arise, predicting collapse in the absence of coercion. Ostrom follows up by demonstrating that such outcomes are not inevitable, submitting an institutional arrangement to enable sustainable governance. However, both frameworks assume a starting point whence the problem is recognized and the relevant actors are engaged.

Carson occupies the space between these two arguments. Her work does not diagnose the commons, nor does it design governance institutions. What *Silent Spring* did was perform the foundational task of making the commons visible. This book transformed dispersed practices into a shared problem, building the foundation that collective action depends upon.

She breaks Hardin’s pessimism by showing that behavior is not fixed and that public understanding can shift in ways that encourage cooperation. At the same time, she provides the preconditions that Ostrom’s framework depends on but does not theorize. Most importantly, Carson operates at a scale that neither framework captures.

The DDT case suggests a revised sequence for commons theory that includes an intermediate stage that includes problem recognition, mobilization, and legitimacy building before moving from dilemma to governance. Hardin identifies the problem; Carson makes it accessible and actionable; Ostrom provides the tools for governing it. This intermediary stage also reveals an underdeveloped dimension in both frameworks, the polycentric activation on a nation scale.

Ostrom's theory of polycentric governance does not fully account for how such systems are activated in the first place, especially in large-scale, borderless, contested contexts.

Recognizing Carson as a bridging mechanism allows for a more complete understanding of commons governance. It shifts the focus from assumptions about behavior or institutions to the conditions under which collective action becomes possible. The following section formalizes these stages into a reconstructed commons governance framework that integrates problem recognition, norm building, and polycentric activation that leads to institutional design.

7. Proposed reconstructed commons governance frameworks

Both Garrett Hardin and Elinor Ostrom begin commons theory from a point where the dilemma is recognized, and actors are in a position to respond. The case of *Silent Spring* shows that this starting point cannot be assumed, especially for large-scale, diffuse, and contested contexts. To address this limitation, a reconstructed commons governance framework that conceptualizes new stages of collective action, from problem recognition to enforcement, is proposed in Figure 2.

Figure 2.

Stage 1: Problem Recognition

- Carson reveals the ecological commons, names the threat, and makes invisible harms visible in *Silent Spring*.
- Communication media creates common knowledge in a broad audience through *The New Yorker* serialization.
- Activates moral norms, asks if new norms are necessary.
- Breaks Hardin's pessimism about inevitability.

Stage 2: Norm Activation & Public Mobilization

- Carson's narrative builds shared norms and common knowledge that others begin to act on.
- Media amplification, public debate, and scientific contestation begin to legitimize environmental concerns and build political pressure.
- Mass mobilizing involves citizens, journalists, scientists, and advocates.
- Absent from both Hardin and Ostrom

Stage 3: Polycentric Activation

- Multiple levels of governance and authority respond to mobilization and political pressure by engaging with solving the issue.
- Participants in this stage include powerful stakeholders, news media, and government agencies.
- Carson triggers the issue getting escalated to the federal government but does not coordinate it escalating at this level.
- New governance emerges through conflict, debate, expertise, and political struggle rather than cooperation or local problem solving.

Stage 4: Institutional Review

- After escalating debates that involve hundreds of news articles and television interviews, *Silent Spring* reaches President Kennedy, who orders a study on pesticides.
- The issue is formally legitimized and institutions begin evaluating the problem and solutions.
- This is where Ostrom's framework begins.

Stage 5: Rule Formation

- EPA and federal agencies craft regulations.
- Actors of authority took over pesticide regulation after Carson passed away.
- Ostrom's principles: monitoring, nested governance partially applies.

Stage 6: Enforcement

- DDT is banned; monitoring continues.
- Coercion through command-and-control regulation.

Reconstructed Commons Governance Framework, Applied to DDT Case

This framework introduces three pre-institutional stages; problem recognition, norm activation, and polycentric activation, which precede and enable the institutional dynamics emphasized in existing theory. By incorporating these stages, problem recognition, norms and mobilization, and polycentric activation, the framework provides a more complete account of how commons are not only governed but made governable. It is designed to apply not only to localized problems, but also to national and large-scale problems that the current frameworks struggle to address.

Stage 1. Problem Recognition: The first stage involves the transformation of a condition into a commons dilemma. In the DDT case, Carson performs this role through the publication of *Silent Spring*. Invisible harms are made visible as part of a shared environmental problem. This stage generates common knowledge, reframes individual actions, and activates moral concern. This new step is fundamental in establishing the conditions under which collective action becomes an option. In doing so, this new framework corrects the structural inevitability argued by Hardin by constructing new, universal understandings. Further, the audience needs to recognize a shared fate to encourage diverse actors forming shared norms.

Stage 2. Norm Activation and Public Mobilization: This stage builds on problem recognition by translating awareness of the problem into engagement with the problem. Carson's narrative was strong enough to mobilize itself. After publishing, *Silent Spring* was amplified by the media, debated publicly, and engaged with by the scientific community (PBS, 2017). Citizens, journalists, scientists, and advocacy groups participate in this stage to shape and reinforce new norms and generate political pressure. This stage is largely absent from both Hardin and Ostrom. While Hardin dismisses this and Ostrom assumes it will happen, neither theorizes how norms emerge at scale. The DDT case shows that norm activation is a dynamic and contest process to align diverse actors around a shared understanding of risk.

Stage 3. Polycentric Activation: The third stage involves the activation of multiple, overlapping centers of influence and authority in response to public mobilization. In the wake of *Silent Spring*, a wide range of actors who wield power and authority engage with the issue. For this stage, new actors like national media organizations, government agencies, heads of state, and powerful stakeholders and advocates engage with the issue. Each contributing to its interpretation and response. No single actor controls the outcome, instead, governance emerges as actors interact with each other across domains. The main difference between actors in stage 2 and 3 is that actors in this stage will have authority and power. This is not to say that actors in stage 2 do not have power or authority, but that the authority is specifically deployed in this step.

Unlike Ostrom's concept of polycentric governance, which emphasized cooperative arrangements among local actors, this stage is characterized by contestation, expertise, and political or rhetorical conflict. The DDT controversy involved industry resistance, scientific disagreement, media framing, personal attacks, and legislative inquiry. This produces an often-adversarial process.

In cases where there is gridlock; those in authority question the legitimacy of the issue or those harmed are not satisfied with the outcomes, these cases will stall, not fail. If the issue languishes in Stage 3 for too long, discourse can become incredibly volatile and reactive. The process becomes circular, re-entering Stage 1 and beginning a recursive cycle through Stage 3, escalating each time, moving more rapidly through each phase as the narrative builds on prior conflict, until authorities finally establish proper norms and legitimacy is gained.

While Carson triggers this activation, she does not theorize how these actors coordinate or compete, and existing commons theory provides limited guidance at this scale. Because of this, polycentric activation represents a critical but underdeveloped stage in understanding large-scale commons theory. This is the last step that Carson participated in and only through a handful of interviews and written replies.

Stage 4: Institutional Review: The fourth stage marks the transition from public discourse to formal evaluation within governance institutions. In the DDT case, this was the review conducted by the Presidential Science Advisory Committee. This stage, as with Carson's case, will validate the claims and the issue will gain formal legitimacy as an object of policy consideration. With the conflict out of the way, formal cases on the matter will be proposed. This is the point where Ostrom's framework begins to apply directly. However, the preceding stages are what make this review possible. When the preceding stages fail to activate or inform actors, environmental policy will languish or fail to address the issue.

Stage 5: Rule Formation: The fifth stage is for the development of formal rules and regulatory frameworks in response to the identified problem. In the DDT case, this was the implementation of pesticide regulations. At this stage, all major parties would have had their say in the matter and the most elements of Ostrom's design principles emerge, particularly monitoring and nested governance.

Importantly, in contrast to existing design principles, rule formation in this context is not the product of localized, cooperative negotiation among resource users, but of political processes shaped by public pressure, scientific expertise, and institutional authority. The local users from Ostrom's framework would have been some of many actors that were involved in Stages 2 and 3.

Stage 6: Enforcement: The final stage involves the implementation and enforcement of rules. In the DDT case, this is the eventual ban of DDT and the establishment of government institutions for monitoring and regulatory oversight of pesticides. While included in existing the commons framework, it was analytically incomplete without the preceding stages. Now there is theoretical reasoning to put enforcement at the end of the collective action process specifically.

Someone, or something, will bring an issue to the public's attention. People will begin mobilize after learning about the issue, new norms will activate, and the public will rally and demand solutions. Next, the issue escalates into the media, experts submit their own opinions, and a competitive, non-centralized debate takes place. At the conclusion of those discussions, formal institutions take over solving the problem and review the facts that have come up before formally proposing new rules and, finally, enforcing them.

More broadly, this framework reorients commons theory from a static analysis of institutions or incentives to a dynamic understanding of how environmental problems are constructed, contested, and ultimately governed. By incorporating pre-institutional processes and recognizing the role of communication, norms, and multi-actor engagement, it provides a more comprehensive model of collective action that is better suited to the complexity and scales of modern environmental challenges.

8. Limits of the Reconstructed Framework

8.1 Theory Limits and Breakdowns

While the reconstructed governance framework helps explain why *Silent Spring* produced a transformative policy response, there are several examples of where theory continues to fall short. Figure 3 summarizes these proposed limitations, failures, and real-world examples.

First, and most nefariously, in Stage 1, there can be coordinated efforts to prevent problem recognition. In the DDT case, Carson was able to overcome smear campaigns, mostly by being reasonable and uncriticizable on the science (PBS, 2017). However, Stage 1 can be actively disrupted by actors who profit off of the proliferation of the issue.

Second, in Stage 2, there are cases where the new norms formed Stage 2 are polarizing or rejected by large portions of actors who are impacted by the problem. If uniform knowledge is not adequately communicated, actors can disagree on norms and shift focus to second-order problems of compliance, rather than the harm or threat.

Finally, in Stage 3, gridlock is possible if authorities disagree on the legitimacy of a problem. Often, these are cases where real-world harm has existed but was not recognized as legitimate or where the harm has persisted and been lived with for a long time. Even when experiencing harm directly, commons recognition can fail without legitimacy. Rather than collapsing, these situations tend to linger in a state of unresolved tension as authorities struggle to impose a solution. Gridlock or insufficient rule making can create a second order commons dilemma that can cycle back through the earlier stages, often with greater intensity, repeating until shared norms are established and legitimacy is recognized.

Figure 3.

Stage	Failure Mode	Examples
Stage 1: Problem Recognition	Open to disruption: Actors who benefit from the problem work to ensure it is proliferated.	Climate change denial
Stage 2: Norm Activation and Public Mobilization	Prone to polarization: Insufficient knowledge or poor communication prevents consensus.	Climate change solutions, COVID masks
Stage 3: Polycentric Activation	Susceptible to gridlock: Institutional fragmentation and adversarial politics prevent convergence. Insufficient rules create second order commons dilemma, restart process from Stage 1.	Operation silver shovel, Love Canal

Limits of Reconstructed Commons Governance Framework

8.2 *An Inconvenient Truth*

The 2006 documentary *An Inconvenient Truth*, presented by Al Gore, sought to do for climate change what *Silent Spring* accomplished for pesticides: transform a complex scientific issue into a recognized problem requiring urgent action. The film combined scientific data, visualizations, personal narratives, and lecture format into a cinematic production. The film was widely seen, received critical acclaim, and contributed to raising awareness of climate change, even winning an academy award for Best Documentary Feature (Guggenheim, 2006). By all measures, the film was highly successful.

However, if evaluated against its broader goal of shifting public understanding and sustained collective action, the outcome is not so clear. Climate change remained politically polarized and there was no regulatory response. Unlike the DDT case, where public mobilization led to rapid and lasting institutional change, the response to *An Inconvenient Truth* was fragmented and uneven. Why did Gore's attempt at problem communication fall short of the transformative impact achieved by Carson?

Stage 1: Problem Recognition, Gore faced a much different starting point than Carson. Carson introduced an unknown issue, while Gore entered into a contested debate. His task was not revealing the commons, but to confront skepticism and align problem recognition on a global scale. The film succeeded in consolidating awareness in viewers that were already receptive, but struggled to establish common knowledge across divided groups, leaving the problem unrecognized.

Stage 2: Norm Activation and Public Mobilization, the contrast between the two communications becomes obvious. Carson's work encouraged moral concern that cut across political and social boundaries. By contrast, Gore filtered his message through his identity as a political figure which ended up polarizing audiences. More importantly, the film's didactic tone, framed as a persuasive lecture rather than a debate, positioned viewers as either convinced or uninformed. Gore did not leave room for viewers to be skeptical. During a time when climate change was considered controversial, Gore did himself no favors trying to win over skeptics or building universal norms with the matter-of-factness it was presented with. In a way, the film entrenched the existing division. Gore was very different from Carson, who, when she sat for the few television interviews after *Silent Spring*, came off as calm and sensible. Carson urged people to gain an understanding of the issue while Gore told people it was urgent to understand the issue now, denying actors the opportunity to come to the conclusion on their own terms. Perhaps Gore would have been better off asking if we are emitting too much CO2 rather than telling us we must emit less.

Stage 3: Polycentric Activation, the differences reflect broader structural conditions. While *An Inconvenient Truth* did engage multiple actors, the responses did not converge together as they did in the DDT case. Instead of producing coordinated escalation, the arguments made by the film entered into an already fragmented discussion space with competing narratives and organized opposition. While Carson dealt with similar opposition and strategic misinformation, a diverse assortment of actors aligned alongside Carson to elevate the message of *Silent Spring* while Gore's efforts were absorbed into an existing landscape that limited the opportunities for polycentric activation.

Without the shared problem recognition and norm activation, the later stages of commons governance could not proceed for Gore with the same level of impact or legitimacy as Carson. The comparison highlights the effectiveness of environmental communication is not determined by the accuracy of information or the scale that the message reaches, but how it interacts with existing social, political, and communication conditions. Carson's success can be credited to her ability to construct a shared understanding of risk using narrative and scientific authority to unify rather than divide audiences. Gore, however, operated in an environment where the problem was already entangled in political identity.

9. Conclusion

This paper begins by asking how environmental problems become commons dilemmas in the first place? By examining the transformation of pesticide use from an unquestioned public good to a widely recognized ecological hazard following the publication of *Silent Spring*, it becomes clear that collective action does not begin with institutions or incentives, but with recognition. The DDT case demonstrates that before a commons can be governed, it must first be constructed as a shared problem that creates a common fate among dispersed actors. This process depends not only on scientific knowledge, but on communication, narrative, and the capacity to inform individuals of complex risks as morally and politically meaningful.

With this in mind, Rachel Carson reveals a foundational dimension of commons governance that both Garrett Hardin and Elinor Ostrom leave underdeveloped and assumed. Hardin's model predicts an inevitable tragedy. Yet the DDT case shows that preferences and perceptions can change the definitions of self-interest as new knowledge updates existing norms. The tragedy is contingent on whether or not actors come to recognize the consequences of inaction. Carson shows how communication and collective action can interrupt the tragedy.

Similarly, while Ostrom provides a powerful framework for understanding how commons can be successfully governed, her design guidelines are biased towards smaller-scale systems where communication, trust, and shared fates already exist and can be assumed to be strong. The DDT case exposes the limits of Ostrom's principles at larger scales. For national scales, the preconditions must be actively constructed. Carson shows us how that construction occurs, not through governance, but through problem recognition, norm formation, and public mobilization that precede current institutional designs. In doing so, she extends commons theory beyond its traditional scope and reveals governance dynamics that operate at the national level.

The reconstructed framework proposed in this paper integrates these insights into a sequential model of collective action. Governance of commons is not a single-stage issue, but a progression of recognition to enforcement. By introducing pre-institutional stages, it provides a more complete account of how environmental issues become governable, particularly in large-scale and contested contexts. This framework not only reconciles key tensions between Hardin and Ostrom but also bridges their theories by situating them within a broader process. Hardin defines what a commons is and what the structural risks are, Carson makes that risk visible and actionable, and Ostrom provides the mechanism for governance once collective action is underway.

More broadly, this analysis highlights the central role of individuals in shaping environmental outcomes. Carson's impact underscores that transformative change can originate not only from institutions, but from those who reframe problems in ways that the public can understand and mobilize behind, creating the conditions for collective action. As environmental problems become more complex and diffuse, solving them wont only rely on better rules, but on fostering knowledge and normative commitments that make those rules possible.

This research can be expanded to consider additional frameworks in commons theory outside of Hardin and Ostrom and how the reconstructed framework performances in global governance systems. The reconstructed framework excels when identifying new commons issues where it was previously invisible but struggles when problems are actively debated, actors disagree on suggested norms, or when mobilized audiences did not achieve satisfactory polycentric activation.

Ultimately, the contribution of this paper is to shift the analytical starting point of commons theory. Rather than assuming that problems are already recognized and actors can respond, it precedes the process through which recognition, common fate, and collective will are constructed. By doing so, it offers more dynamic and scalable understandings in commons governance. One that accounts for how tragedies can be interrupted, how cooperation can emerge, and impossible problems become governable commons.

References

Carson, R. (1962). *Silent Spring*. Mariner Books.

Guggenheim, D. (Director). (2006). *An Inconvenient Truth* [Film]. Lawrence Bender Productions.

Hardin, G. (1968). The Tragedy of the Commons. *Science, New Series*, *162*(3859), 1243–1248.

Ostrom, E. (2000). Collective Action and the Evolution of Social Norms. *Journal of Economic Perspectives*, *14*(3), 137–158.

PBS, A. E. (Director). (2017, June 20). *Rachel Carson | Full documentary | AMERICAN EXPERIENCE | PBS* [Video]. YouTube. https://www.youtube.com/watch?v=mFDh9c34XX4

Appendix

Table 1. *Evaluation of Harden's Assumptions from Tragedy of the Commons Within the Context of the DDT Case Study*

Hardin's Assumption	**Hardin's Meaning**	**Carson / DDT Case**	**Fit**	**Explanation**
1. Rational Self-Interest Drives Behavior	Individuals will maximize personal gain, even if it leads to collective ruin.	Farmers and chemical companies initially acted in line with self-interest (higher yields, profit). However, once Carson revealed ecological harm, many citizens supported regulation that limited immediate benefits.	Partial	Carson confirms Hardin at the level of initial behavior but shows that preferences are not fixed. Once people understand collective risk, they can support restraint, contradicting Hardin's static model of self-interest.
2. The Commons Leads to Inevitable Overuse (Tragedy)	Hardin insists that degradation is structurally inevitable	DDT use did produce ecological harm (bioaccumulation, species decline), consistent with a tragedy dynamic. However, public mobilization led to regulation and reversal.	Partial (Diagnosis strong, prediction weak)	Carson validates Hardin's diagnosis of ecological risk but disproves inevitability. The tragedy is not unavoidable. It can be interrupted through social and political processes.
3. Lack of Trust and Norms Prevent Cooperation	Hardin assumes that individuals cannot rely on others to restrain themselves, making voluntary cooperation unstable.	Carson actively builds shared norms (stewardship, responsibility) and trust in scientific knowledge. Public concern becomes widespread and coordinated.	Weak	Carson demonstrates that trust and norms can be constructed through communication. Hardin assumes their absence; Carson shows they can emerge and enable collective restraint.
4. Communication Is Irrelevant or Ineffective	Hardin explicitly rejects appeals to conscience or persuasion	*Silent Spring* fundamentally changes public perception, turning an invisible problem into a shared concern and catalyzing action.	Very Weak	Carson directly contradicts this assumption. Communication is decisive. Without it, the commons problem remains invisible and unaddressed.
5. Information Does Not Change Outcomes	Knowledge cannot overcome incentives, even when people understand the problem	Before Carson, the public lacked knowledge of DDT's harm. After widespread dissemination of information, behavior and political pressure changed significantly.	Weak	Carson shows that information is transformative. Knowledge reshapes preferences and informs what responses are rational, making collective solutions viable.
6. The Public Is Passive	The public is structurally unable to self-organize.	Carson's work mobilizes citizens, scientists, journalists, and advocacy	Weak	Carson demonstrates that the public can be highly active when properly

	Only coercive structures can solve commons problems.	groups. Public pressure drives congressional hearings and regulatory change.		informed and engaged. This challenges Hardin's assumption of passivity.
7. Only Coercion or Privatization Can Solve the Problem	Mutal coercion, mutually agreed upon, is the only viable solution. Restraint is impossible.	The DDT ban is ultimately a form of state regulation (coercion), but it emerges from democratic mobilization rather than imposed authority alone.	Partial	Carson partially supports Hardin in that regulation is required, but she shows that coercion is politically constructed through public engagement, not simply imposed from above. It is the result of collective action, not the primary solution.
8. Power and Politics Are Secondary	Commons dilemmas are structural and apolitical, focused on incentives, not institutions	The DDT controversy involved intense conflict: industry resistance, scientific debate, media battles, and government intervention.	Weak	Carson reveals that commons problems are deeply political. Power, conflict, and legitimacy are central, elements missing from Hardin's model.

Note: This table assesses how the DDT case fits within Hardin's assumptions about the tragedy of the commons. Adapted from (Carson, 1962; Hardin, 1968)

Table 2. *Evaluation of Ostrom's Design Principles Within the Context of the DDT Case Study*

Ostrom's Design Principle	**Ostrom's meaning:**	**Carson / DDT Case**	**Fit**	**Explanation**
1. Clearly Defined Boundaries	Successful commons require clear boundaries around who uses the resource and what the resource is.	The ecological systems Carson describes (air, water, soil, food webs) are boundaryless. DDT travels across farms, counties, states, and continents. No community of "users" can be clearly identified; the harms are diffuse and involuntary.	Weak	Carson exposes a type of commons (planetary ecological systems) that cannot be bound in Ostrom's framework. global, borderless harms are a limit of Ostrom's design principles for collective action.
2. Rules Matched to Local Conditions	Rules should be tailored to the specific ecological and social context.	The eventual DDT ban was national, not locally tailored. The ecological processes Carson describes (bioaccumulation, persistence) are not local; they are systemic. Local variation in pesticide uses mattered, but the harms were non-local.	Partial	Carson shows that some environmental problems cannot be governed by local rule variation because the ecological processes themselves are not local. This reveals a scale mismatch in Ostrom's model.
3. Collective-Choice Arrangements	Users of the resource help make the rules.	Farmers, chemical companies, and affected citizens did not co-design pesticide rules. The regulatory response was top-down (EPA, federal ban). Carson's strategy was public mobilization, not user negotiation.	Weak	The DDT case is a story of public pressure. State regulation, not self-governance by appropriators. Carson reveals that some commons problems require political conflict, not cooperative rulemaking.
4. Monitoring by or Accountable to Users	Users monitor resource conditions and each other.	Carson herself functions as a monitor, aggregating scientific evidence. Citizens become monitors through her narrative; they begin noticing bird deaths, fish kills, etc. But monitoring is not done by "users"; it is done by scientists, journalists, and the public.	Strong but indirect	Carson strengthens the monitoring function by making invisible harms visible. She shows that monitoring can be scientific and communicative, not just user based.

5. Graduated Sanctions	Rule violators face escalating, proportionate sanctions.	The DDT ban is binary: legal → illegal. No graduated sanctions, no negotiated penalties. Enforcement is adversarial and regulatory, not communal. Also, Carson did not call for outright bans, only informed and moderated usage.	Weak	Carson's case shows that for large-scale industrial harms; graduated sanctions are irrelevant; State coercion was the solution in DDT. This exposes a limit in Ostrom's model for industrialized societies.
6. Accessible Conflict Resolution	Users need accessible ways to resolve disputes.	Conflict was high cost: congressional hearings, lawsuits, public attacks on Carson, scientific panels. Chemical companies fought aggressively. There was no shared forum for resolution.	Weak	The DDT case demonstrates that some environmental conflicts are structurally adversarial and cannot be resolved through low-cost mechanisms. This is another limit of Ostrom's model for high-power asymmetries.
7. Recognition of the Right to Self-Organize	External authorities must allow communities to form their own governance institutions.	What is recognized here is not a "user community" but the public's right to mobilize. Citizens, scientists, and NGOs gained legitimacy to challenge chemical use. The state recognized environmental advocacy, not self-governance.	Partial	Carson shows a different kind of "self-organization": public-interest mobilization, not resource-user governance. This expands Ostrom's concept beyond local appropriators.
8. Nested Governance	Large-scale commons require multi-level governance (local → regional → national).	This is where Carson fits best. Local concerns → state hearings → Presidential Science Advisory Committee → EPA → national ban. The governance response is clearly multi-level.	Strong	Carson catalyzes a nested governance response, even though she herself is not designing institutions. This is the principle where Ostrom and Carson most clearly overlap.

Note: This table assesses how the DDT case fits within Ostrom's design principles for commons governance. Apapted from (Carson, 1962; Ostrom, 2000)